1. **List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.**

IP Class	Address Range	Example
Class A	1.0.0.0 – 126.255.255.255	Large company network
Class B	128.0.0.0 – 191.255.255.255	University/college network
Class C	192.0.0.0 – 223.255.255.255	Home Wi-Fi router
Class D	224.0.0.0 – 239.255.255.255	Video streaming/multicast
Class E	240.0.0.0 – 255.255.255.255	Experimental/research use

2. **Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses. show your working steps for each calculation.**

Given:

- **IP Address: 192.168.10.45**
- **Subnet Mask: 255.255.255.0 (/24)**

Step 1: Network Address

With /24, the first 3 octets are the network part.

Network Address = 192.168.10.0

Step 2: Broadcast Address

The last octet can go from 0 to 255.

Broadcast Address = 192.168.10.255

Step 3: Valid Host Range

- First usable = Network + 1 → **192.168.10.1**
- Last usable = Broadcast − 1 → **192.168.10.254**

Item	**Address**
Network Address	192.168.10.0
First Host	192.168.10.1
Last Host	192.168.10.254
Broadcast Address	192.168.10.255
Valid Hosts	**254**

3. **Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.**

- **Using 10.0.0.0/24 and needing 4 separate subnets, we can divide it into 4 equal /26 subnets.**

Group	Network Address	Subnet Mask	Usable Host Range
Delivery Partners	10.0.0.0	255.255.255.192 (/26)	10.0.0.1 – 10.0.0.62
Restaurants	10.0.0.64	255.255.255.192 (/26)	10.0.0.65 – 10.0.0.126
Customers	10.0.0.128	255.255.255.192 (/26)	10.0.0.129 – 10.0.0.190
Admins	10.0.0.192	255.255.255.192 (/26)	10.0.0.193 – 10.0.0.254

4. **You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet. Constraint: Show your calculation steps and final answers.**

Given:

- **Network: 172.16.0.0/16**
- **Required subnets: At least 20**

➢ **Step 1: Find borrowed bits**

We need:

$2^n \geq 20$

- $2^4 = 16$
- $2^5 = 32$

So, 5 bits are borrowed.

➢ **Step 2: New subnet mask**

Original: **/16**
Borrowed: **5 bits**

New prefix = 16 + 5 = /21

Subnet mask:

255.255.248.0

- **Step 3: Hosts per subnet**

Remaining host bits:

32 – 21 = 11 bits

Total addresses:

2^{11} = 2048

Usable hosts:

2048 – 2 = 2046

Item	**Answer**
Original Network	172.16.0.0/16
Required Subnets	At least 20
Bits Borrowed	**5 bits**
Number of Subnets	**32**
New Subnet Mask	**255.255.248.0 (/21)**
Usable Hosts/Subnet	**2046**

5. **A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.**

- The IP address **192.168.1.300** is invalid because each part of an IPv4 address must be between **0 and 255**. The value **300** is outside this range.

For the **192.168.1.0/24** subnet, a valid host IP could be **192.168.1.100** or **192.168.1.200**.

So, the device should be given a valid IP address where the last number is between **1 and 254**.